School of Physics, Engineering and Technology

# BEng Project Initial Report

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**Project Title:** Structured Riemannian Pruning of Feedback Delay Networks

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**Abstract**

Artificial reverberation remains constrained by a persistent quality–efficiency gap. Measured Room Impulse Responses (RIRs) provide high acoustic fidelity, yet low-latency convolution is expensive when many long responses must be instantiated concurrently. Feedback Delay Networks (FDNs) offer a more attractive recursive alternative, but the acoustic smoothness required for colourless late reverberation increasingly favours very high orders, at which the dense feedback matrix introduces an $O(N^2)$ per-sample mixing bottleneck. This project investigates whether such massive, lossless FDNs can be systematically reduced to hardware-efficient structure without sacrificing either stability or perceptual quality.

A direct application of conventional unstructured pruning is unsuitable. In the lossless setting, the feedback matrix is constrained by $A^T A = I$, so arbitrary element removal generally violates the orthogonality constraint and destroys the very geometry that guarantees bounded energy. Moreover, the sparse patterns produced by magnitude pruning are poorly aligned with single-instruction multiple-data execution and cache-friendly memory access.

The proposed solution is therefore not to prune entries independently. Instead, the project starts from a dense lossless reference operator, either learned through FLAMO's implicit orthogonal parameterisation or specified directly at high order, and then projects it onto a structured feasible subset centred on block-circulant topology. The reduction step is guided by ranking criteria derived from temporal activations and perceptually weighted response error rather than raw coefficient magnitude.

The resulting structure will be exported as a deployment blueprint and realised as a custom C++ inference node using RTNeural-style real-time design and AVX2 vectorisation. The longer-term goal is to convert dense differentiable FDNs into real-time viable reverberation engines.

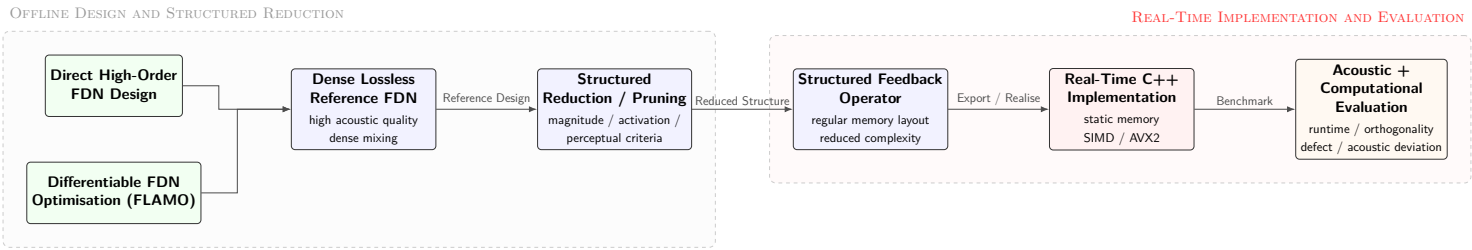


**Figure 1:** Overall project pipeline. A dense acoustically effective FDN is first obtained either by direct high-order construction or by differentiable optimisation in FLAMO, then reduced to a structured form and finally evaluated in real-time implementation.

# 1 Introduction and Aims

**Aim.** The aim of this project is to develop and evaluate a structured pruning method for dense lossless Feedback Delay Networks, such that acoustically effective high-order designs can be reduced to computationally efficient real-time forms without unacceptable loss of stability or sound quality.

**Objectives.** To support this aim, the project has four objectives:

1. construct dense lossless FDN reference systems for high-order diffusion and RIR-matching scenarios, and verify their baseline acoustic behaviour and stability;
2. develop at least one structured reduction method for mapping dense feedback operators onto a computationally efficient topology, and quantify the resulting structural and orthogonality error;
3. compare magnitude-, activation-, and perceptually guided reduction criteria using defined acoustic and computational metrics;
4. implement a real-time C++ prototype of the structured design and benchmark it against the dense baseline in terms of runtime, memory footprint, orthogonality defect or approximation error, and acoustic deviation.

Artificial reverberation has long been shaped by a fundamental trade-off between acoustic realism and computational efficiency, often described in digital signal processing (DSP) as the quality-efficiency gap. At one end of this continuum, convolution with measured Room Impulse Responses (RIRs) provides a highly faithful representation of a physical space. However, the need to process long sample sequences results in a complexity that scales as $O(L \log L)$, where $L$ denotes the impulse-response length. Even when fast convolution methods are employed, the associated memory traffic, buffering overhead, and latency penalties remain substantial, making convolution computationally prohibitive for multiple concurrent instances under low-latency real-time constraints on consumer-grade central processing units (CPUs).

By contrast, algorithmic reverberators, most notably Feedback Delay Networks (FDNs), offer computationally efficient recursive structures whose per-sample cost scales approximately linearly with the number of delay lines, $N$. In such systems, reverberation is generated not by

direct reproduction of a measured impulse response, but through recirculation within a network of delay lines coupled by a feedback matrix. This architecture has made FDNs one of the most influential formulations in algorithmic reverberation, since they provide a favourable balance between computational efficiency, parametric flexibility, and perceptual plausibility. Yet traditional low-order FDNs, typically based on $4 \times 4$ or $8 \times 8$ feedback matrices, often exhibit insufficient modal density. The resulting sparsity of resonant modes gives rise to metallic coloration and audible ringing artefacts, since the auditory system can resolve individual resonances that do not blend into a diffuse statistical sound field.

A natural response is therefore to increase the order of the network. In principle, high-order FDNs can produce denser mode distributions, faster echo build-up, and more nearly diffuse late reverberation. However, this improvement is not obtained without cost. If the feedback matrix is dense, the mixing stage scales quadratically, so that the per-sample cost of matrix-vector multiplication becomes $O(N^2)$. For sufficiently large networks, this mixer dominates the computational budget, even if the delay lines themselves remain inexpensive. The central difficulty is thus to retain the acoustic benefits of high-order, lossless mixing without inheriting the full computational cost of a dense feedback operator. In practical terms, the project asks whether a large artificial reverberator that already sounds good can be simplified into a much faster structure without losing too much of its acoustic character.

This project argues that the appropriate response to this difficulty is not merely better optimisation, but structured pruning. In the present context, pruning denotes the systematic reduction of a dense Feedback Delay Network into a computationally efficient structured form by removing, constraining, or reorganising degrees of freedom in its feedback matrix while preserving, as far as possible, the stability, diffuseness, and acoustic behaviour of the original design. Although pruning is well established in neural-network compression as a means of reducing computational cost while retaining useful global behaviour, it has not become a standard design principle for Feedback Delay Networks. The present work therefore examines whether that idea can be reformulated for artificial reverberation, where the objects being pruned are not arbitrary parameter arrays but lossless feedback operators subject to geometric, dynamical, and perceptual constraints.

The intended contribution of the project is a structured, geometry-aware reduction method for dense lossless Feedback Delay Networks, that is, for reducing dense feedback structures while respecting the geometry that keeps them stable. The term *Riemannian* is essential here because the feedback matrices of interest must remain within constrained classes such as orthogonal or otherwise lossless operators. Pruning therefore cannot be treated as unconstrained coefficient removal alone. Instead, the reduction process must respect the geometry of the feasible set. The novelty of the work lies not simply in applying manifold-aware optimisation to FDNs, nor simply in borrowing pruning language from machine learning, but in combining the two into a single methodology for converting dense, acoustically effective FDNs into efficient structured real-time systems.

Two application settings motivate this framework. The first is the construction of extremely

high-order FDNs, for example with $N = 1024$ or $N = 4096$, whose dense mixing would otherwise be too expensive for practical use. In this regime, the goal of pruning is to preserve the diffuse, high-modal-density behaviour needed to suppress metallic artefacts while approaching the efficiency of fast structured transforms. The second is the deployment of differentiable reverberators, particularly in frameworks such as FLAMO, where dense orthogonal feedback matrices may be optimised to mimic a target Room Impulse Response. In that setting, pruning is intended to transform a learned but computationally heavy FDN into a real-time implementable structure while preserving acoustic fidelity to the reference RIR.

For an electronic engineer, the essential point is simple: a lossless feedback matrix cannot be edited coefficient by coefficient in the same way as an ordinary unconstrained gain matrix, because its entries are coupled by the energy-preserving condition.

## 1.1 Why the feedback matrix is a geometric object

For a real-valued lossless FDN, the feedback matrix must preserve Euclidean energy, and therefore satisfy

$$A^T A = I.$$

The feasible set is therefore not the full ambient space $\mathbb{R}^{N\times N}$ but the orthogonal group $O(N)$, i.e. the square case of the Stiefel manifold [1]. This has an immediate engineering consequence: once orthogonality is imposed, the coefficients are no longer free to vary independently. Admissible perturbations must satisfy

$$X^T V + V^T X = 0,$$

so the degrees of freedom are coupled by the constraint itself [1].

This coupling matters directly for pruning. If the feedback matrix is treated as just a table of coefficients, then deleting or modifying individual entries appears harmless. Geometrically, however, such edits generally leave the feasible set immediately. In this setting, pruning cannot be understood as independent coefficient removal. The feedback matrix is a geometric object, and any principled reduction procedure must respect that geometry [1].

## 1.2 Why unstructured magnitude pruning fails in this setting

In neural-network pruning, the canonical baseline is iterative magnitude pruning. This consists of training a dense model, removing the parameters of smallest magnitude, and optionally retraining the surviving subnetwork [5]. This idea is historically important and often empirically effective in over-parameterised learning systems. It is therefore a natural point of comparison. However, the assumptions that make magnitude pruning plausible in generic neural networks do not transfer cleanly to lossless FDN mixing.

The first difficulty is geometric. In an orthogonal feedback matrix, coefficient values are not independently meaningful in the usual way, because row and column norms are globally constrained. Magnitude is therefore a poor proxy for importance. In canonical dense orthogonal

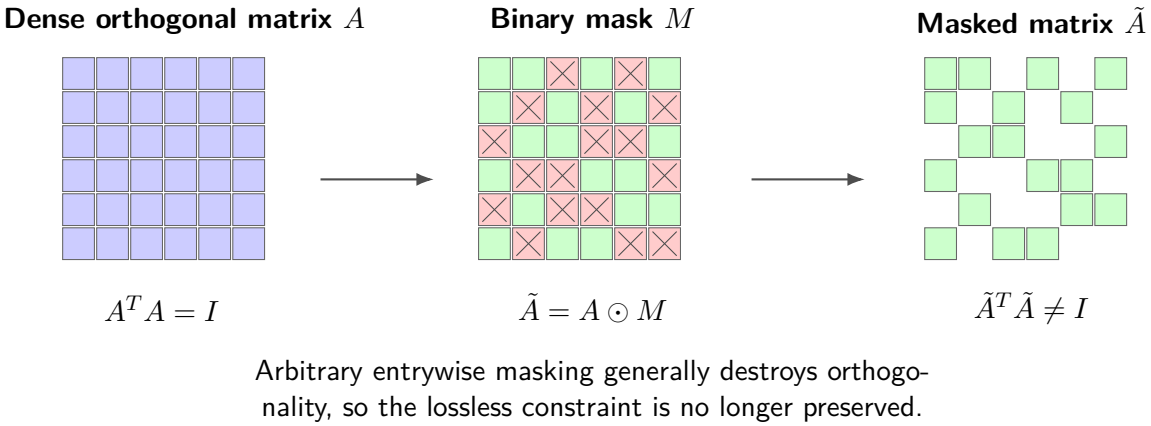


**Figure 2:** Schematic illustration of why naive entrywise pruning fails for lossless feedback matrices. Arbitrary masking of an orthogonal matrix generally destroys the orthogonality condition and therefore breaks the lossless constraint.

constructions such as the normalised Hadamard matrix, every entry has identical magnitude $1/\sqrt{N}$, so there is no intrinsic population of "small" coefficients to remove. More generally, even when magnitudes are unequal, the effect of a single coefficient is mediated by the orthogonality of the whole operator. Removing one entry does not merely delete a weak connection; it perturbs a balance condition shared by the corresponding row and column.

The second difficulty is feasibility. If $A$ is orthogonal and one forms a masked matrix $\tilde{A} = A \odot M$, with $M$ a binary mask, then in general $\tilde{A}^T \tilde{A} \neq I$.

Figure 2 illustrates schematically why arbitrary entrywise pruning breaks the lossless constraint.

The lossless property is therefore destroyed by arbitrary entrywise pruning. Any subsequent attempt to restore orthogonality must include some projection or reparameterisation step, at which point the original unstructured pruning decision has already been altered. In other words, naïve sparsification and geometric feasibility are in tension from the outset.

The third difficulty is computational rather than geometric. Unstructured sparsity is not automatically fast. Clarke and Chowdhury point out that sparse matrix multiplies often rely on indirect memory accesses, and that this is especially problematic in real-time audio, where efficient inference depends heavily on hardware-level primitives requiring sequential memory access, such as SIMD operations [3]. This observation is highly relevant here. Even if approximate orthogonality could be maintained under entrywise pruning, the resulting sparsity pattern would still be poorly aligned with the requirements of vectorisation-friendly CPU inference.

For these reasons, unstructured magnitude pruning is best treated here as a useful contrast case rather than as the main design principle. Frankle clarifies why it became the dominant pruning baseline in machine learning [5]; Boumal explains why orthogonality couples the coefficients globally [1]; and Clarke and Chowdhury show why inference-time efficiency depends on structured memory layout rather than sparsity alone [3]. Taken together, these observations justify the shift from unstructured magnitude pruning toward structured, geometry-aware reduction.

At an engineering level, this becomes a question of implementation: should the orthogonality constraint be enforced directly during optimisation, or built into the parameterisation from the outset?

## 1.3 Implicit versus explicit manifold optimisation

From a geometric point of view, direct optimisation on $O(N)$ is perfectly natural [1]. However, the training phase need not be implemented in that explicit form. FLAMO already provides a differentiable environment for frequency-domain optimisation of linear time-invariant audio systems within PyTorch, including orthogonality-preserving matrix parameterisations [6]. In its FDN case study, the feedback matrix is implemented as a learnable matrix with orthogonal mapping, allowing the optimisation of a lossless FDN directly within the differentiable framework [6].

For the present project, this makes FLAMO the practical training vehicle. Boumal remains the right source for understanding the geometry of the feasible set [1], whereas FLAMO provides an implementation pathway for training dense lossless FDNs inside an autodiff system [6]. Thus explicit manifold optimisation remains conceptually central, even if the optimisation procedure is carried out through an implicit orthogonality-preserving parameterisation.

## 1.4 Differentiable DSP and deployment tension

The broader context for this project is differentiable digital signal processing. DDSP argues for combining classic signal-processing structure with automatic differentiation, so that strong inductive biases and interpretable modules can be retained without giving up the expressive advantages of neural learning [4]. FLAMO extends this general philosophy to frequency-domain optimisation of differentiable LTI audio modules, including reverberation structures [6]. This makes it natural to learn a dense, acoustically meaningful FDN in a differentiable environment before asking how it should be deployed.

The deployment problem is governed by a different set of constraints. Training favours flexible tensor abstractions, rich parameterisations, and batch-oriented computation. Real-time audio does not. RTNeural makes this distinction especially clear. In hard real-time systems, one must avoid unbounded-time operations in the audio thread, including memory allocation, thread locking, garbage collection, and similar mechanisms [2]. The same paper argues that, for the small models typical of real-time use, CPU inference with SIMD support is often as fast as or faster than GPU execution, precisely because it avoids the overheads that real-time systems cannot tolerate [2]. Its compile-time API is relevant for the same reason: memory is allocated statically, functions may be inlined, and loops can be unrolled when dimensions are known at compile time [2].

Although RTNeural is a neural inference library rather than an FDN library, the deployment lesson transfers directly. A real-time FDN implementation should favour fixed dimensions where possible, pre-allocated buffers, contiguous memory, bounded-time operations, and arithmetic

that is friendly to SIMD vectorisation. This is precisely why structured pruning matters in the present setting. The goal is not only to reduce parameter count in an abstract sense, but to transform a dense differentiably trained FDN into a form whose topology is compatible with real-time CPU execution.

### 1.5 Critical synthesis

The literature therefore points toward a fairly specific methodological direction. Boumal provides the geometric language needed to formulate the feedback matrix as a constrained object on the orthogonal group rather than as an unconstrained parameter array [1]. Frankle provides the historical and methodological background for pruning, particularly the role of magnitude pruning as a dominant baseline in neural-network compression [5]. Clarke and Chowdhury shift that discussion toward inference-time structure, showing that the difference between sparsity and speed is especially important in real-time audio [3]. DDSP supplies the broader differentiable-signal-processing rationale [4], FLAMO provides a practical framework for training dense orthogonality-constrained FDNs [6], and RTNeural clarifies the systems-level discipline required for deployable real-time C++ implementations [2].

Taken together, the literature motivates the present project. None of the cited sources provides a complete procedure for taking a dense, acoustically effective, lossless FDN and converting it into a structured real-time implementation without unacceptable acoustic loss.

## 2 Intended Approach and Methodology

The methodology is organised as a three-phase pipeline that moves from dense reference construction to structured reduction and finally to real-time deployment.

### 2.1 Phase 1: construction of the dense reference system

The first phase constructs the dense FDN reference system against which later structured reductions will be judged. Two use-cases are considered. The first is a high-order diffuse late-reverberation setting, in which a large lossless FDN is specified directly and treated as the acoustic reference object. The second is an RIR-matching setting, in which a differentiable FDN is optimised in FLAMO to approximate a target room response. In both cases, the aim is to begin from a dense acoustically effective reference before introducing computational structure.

This phase is intentionally dense. Its purpose is to separate acoustical design from computational simplification. If structure is imposed too early, it becomes difficult to determine whether a failure is due to limited topology or inadequate training. The dense reference therefore serves as an upper bound on the quality attainable under the chosen loss, optimisation procedure, and parameterisation.

The output of this phase is a dense reference feedback operator $A_{\text{dense}}$, together with the associated delay and attenuation parameters. Additional analysis will include echo-density

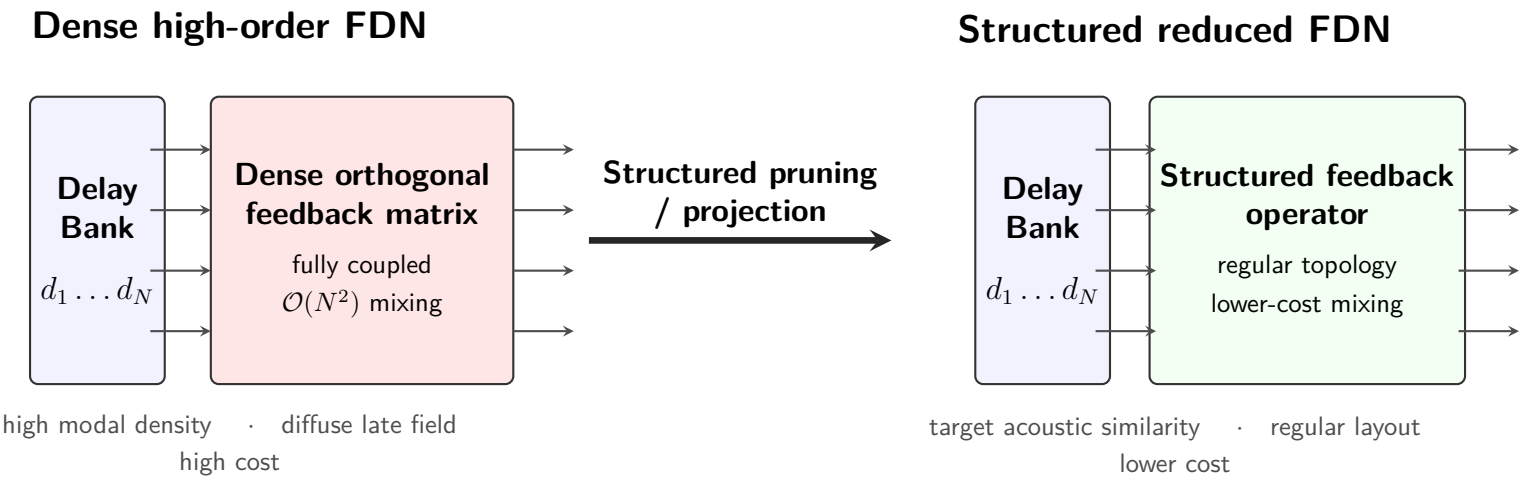


**Figure 3:** Conceptual transition from a dense high-order FDN to a structured reduced FDN.

profiles, spectrograms, and, where applicable, deviation from the target response.

## 2.2 Phase 2: structured projection and psychoacoustic ranking

The second phase forms the scientific core of the project. Here, a dense orthogonal solution is reduced to a structured feasible subset. Block-circulant structure is the primary candidate, since it offers regular memory layout and admits fast multiplication through smaller transform-like operations.

Figure 3 shows the conceptual transition from dense high-order mixing to a structured reduced topology.

At a high level, if $A_{\text{dense}}$ denotes the learned reference, the structured problem is

$$A_{\text{struct}} = \arg\min_{A \in \mathcal{S}} \mathcal{J}(A; A_{\text{dense}}),$$

where $\mathcal{S}$ denotes a structured feasible set, ideally intersecting the orthogonal group or else approximating it closely, and $\mathcal{J}$ measures functional and acoustic deviation.

Three ranking principles will be compared.

**(i) Magnitude baseline.** This is included only as a control. It follows the conventional idea of removing small coefficients or small structural blocks, and is expected to perform poorly for the geometric and computational reasons established earlier.

**(ii) Temporal mean activations.** Inspired by Clarke and Chowdhury's activation-based ranking, the FDN will be driven by representative signals and its internal state trajectories will be analysed. Delay-line couplings or structural blocks whose removal induces only a small change in average propagated activation energy will receive low importance scores. The point is to infer importance from dynamical participation rather than from static coefficient magnitude.

**(iii) Perceptually weighted response error.** A third ranking criterion will be explored based on the deviation between the dense and structured FDN responses in a perceptually weighted time-frequency representation. The guiding idea is that pruning decisions should be judged not only by coefficient values or internal activations, but also by their effect on

acoustically salient properties such as early echo build-up, decay smoothness, and spectral coloration. This criterion is exploratory, and its precise form remains part of the methodological development of the project.

The projection itself may be implemented in stages. One possibility is block selection followed by orthogonal re-fitting within the retained topology. Another is alternating minimisation between a structural mask and a subsequent re-orthogonalisation or structured factorisation step. If exact orthogonality cannot be maintained within the chosen topology, the report will distinguish clearly between exact feasible reduction and approximate reduction. The deviation will then be quantified through metrics such as $\|A^T A - I\|_F$ alongside the acoustic criteria. This distinction matters, because the term "submanifold" is justified only when the feasible set is smooth and closed under the chosen parameterisation; some structured subsets may instead be better understood as discrete topologies equipped with local continuous parameters.

## 2.3 Phase 3: JSON export and real-time C++ realisation

Once a structured model has been obtained, the final phase concerns deployment. Following the general workflow used in RTNeural, the trained or projected parameters will be exported as a compact blueprint, for example in JSON form, and used to instantiate a custom C++ block-circulant node. The implementation will be designed around hard real-time constraints: no dynamic allocation in the audio thread, fixed-size or pre-allocated buffers, contiguous block storage, AVX2-aware vectorised inner loops, and reproducible benchmarking against the dense baseline.

The target computational improvement is to replace naive dense mixing with a structure approaching $O(N \log N)$ arithmetic cost, or at least to achieve a substantially lower constant factor through regular block reuse and vectorisation. That asymptotic claim must, however, be interpreted carefully. It is valid only if the selected structure genuinely admits fast transform-based multiplication. If the final feasible subset deviates from pure circulant algebra, the project will report the measured complexity and cycle counts directly rather than retaining the asymptotic claim by assumption.

Evaluation in this phase will include wall-clock real-time factor, per-block processing time, memory footprint, orthogonality defect where relevant, and acoustic deviation from the dense reference.

In practical terms, the minimum successful outcome of the project will be a working structured reduction and C++ implementation for at least one dense reference FDN, together with quantitative comparison against the dense baseline. The evaluation will focus primarily on FDN orders in the range $N = 64$ to $N = 1024$, with higher orders explored if time permits. A meaningful result would consist of a measurable runtime reduction relative to dense mixing, bounded orthogonality defect or clearly reported approximation error, and no severe reintroduction of metallic artefacts in the reduced design. If the block-circulant route proves too restrictive, the fallback deliverable will be a structured approximation method with explicit trade-off analysis rather than exact manifold-preserving reduction.

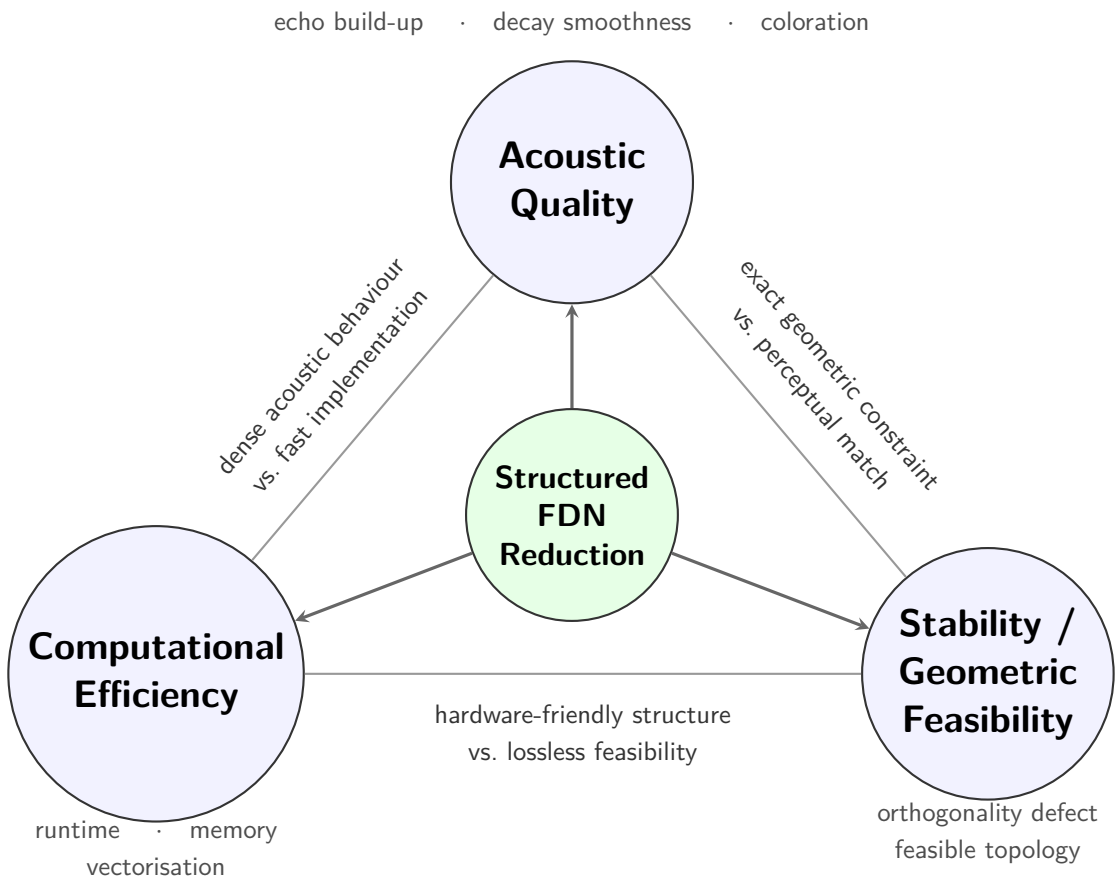


**Figure 4:** Evaluation trade-off for structured FDN reduction. Practical success depends on balancing computational efficiency, geometric or stability-related feasibility, and acoustic quality rather than optimising any one criterion in isolation.

Figure 4 summarises the three-way trade-off that governs the evaluation of structured FDN reduction.

# 3 Project Plan and Timetable

Table 1 summarises the remaining ten weeks of planned project work and maps them onto the objectives and methodology described above.

This schedule mirrors the logic of the project while remaining flexible enough to accommodate technical uncertainty. Objective 1 is concentrated in Weeks 1–2, Objective 2 in Weeks 3–4, Objective 3 in Weeks 5–6, and Objective 4 in Weeks 7–9, with Week 10 reserved for synthesis and report integration. The timetable will be used adaptively throughout the project. If one line of structured reduction proves less viable than expected, effort will be reallocated toward the alternatives that offer the strongest combination of feasibility, efficiency, and acoustic performance. Viva preparation will follow the main body of project work rather than forming part of the core execution schedule.

# 4 Risk Assessment Matrix

**Table 1:** Ten-week project timetable mapped to the intended methodology.

| Weeks | Focus | Planned outputs |
|---|---|---|
| 1–2 | Dense reference construction | Establish dense reference FDN systems for high-order diffusion and RIR matching; verify stability and orthogonality-preserving parameterisation where applicable; define baseline acoustic criteria and collect dense reference responses. |
| 3–4 | Structured reduction design | Investigate feasible structured feedback classes; develop projection, refitting, or reduction strategies from dense orthogonal solutions to structured candidates; implement diagnostics for orthogonality defect and structural approximation error. |
| 5–6 | Pruning criteria and comparative evaluation | Implement and compare candidate ranking or reduction criteria under acoustic, structural, and computational metrics. |
| 7–8 | Real-time implementation | Export structured models and develop an efficient C++ implementation guided by real-time constraints, including static or pre-allocated memory usage and vectorisation where appropriate; validate numerical behaviour against the Python prototype. |
| 9 | Benchmarking | Measure runtime behaviour, memory footprint, and acoustic deviation; compare dense and structured implementations across multiple FDN orders. |
| 10 | Results integration | Consolidate results into tables, figures, and critical discussion; assess the practical trade-off between structural feasibility, stability, and acoustic quality; integrate final findings into the written report. |

**Table 2:** Critical engineering risks and mitigation strategies.

| Risk | Impact | Mitigation |
|---|---|---|
| Structured projection or re-orthogonalisation becomes too expensive for large $N$. | High | Prototype first on moderate orders; separate offline projection cost from online runtime cost; test approximate refitting methods before more expensive projection procedures; cache intermediate results during parameter sweeps where possible. |
| Real-time C++ implementation fails to deliver meaningful speed-up over the dense or scalar baseline. | High | Use aligned storage and contiguous block layout from the outset; maintain scalar reference kernels for parity testing; benchmark micro-kernels independently before full integration; fall back to smaller block sizes or simpler structures if cache pressure dominates. |
| FLAMO optimisation fails to converge to acoustically satisfactory dense references. | Medium to High | Begin from validated FLAMO examples; keep dense training objectives simple before increasing ambition; monitor loss traces, orthogonality defect, and impulse responses; retain fixed analytical baselines for comparison. |
| Projected structured matrices improve efficiency but audibly reintroduce metallic ringing or poor decay behaviour. | High | Compare candidate reductions using acoustic as well as computational metrics; evaluate with multiple signals and decay measures; define minimum acceptable acoustic criteria before accepting speed gains. |
| Exact orthogonality within the chosen structured topology proves too restrictive. | Medium | Compare strict-feasible and approximate-feasible variants explicitly; report orthogonality defect as a first-class metric; if necessary, reformulate the contribution in terms of structured approximation to a dense lossless reference rather than exact manifold-preserving pruning. |

# 5 Ethical Considerations

This project is concerned with mathematical modelling, algorithm design, and real-time software implementation for artificial reverberation. It does not involve human participants, personal data, animal subjects, or safety-critical experimental procedures. No formal ethical ap-

proval is therefore expected to be required. If the scope of the project changes during implementation, for example through the use of external recordings or user evaluation, the relevant university guidance and approval procedures will be followed before that work is undertaken.

# 6 Conclusion and Summary

The work set out in this initial report proposes an approach for taking a dense, acoustically effective lossless FDN, whether obtained by differentiable optimisation or by direct high-order design, and reducing it to a faster structured form suitable for real-time implementation. The project brings together structured reduction, geometric constraint, and deployment-aware implementation within a single methodology.

Even if exact manifold-preserving projection proves too restrictive, the project should still establish something useful. It should indicate how much of the dense orthogonal design can be traded for structured computation before the resulting reverberator ceases to behave like the system from which it was derived.

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